

FIM 590 Algo. and HFT

Homework #1

Solution

1. **Finite-Horizon Quadratic Control.** Define

$$H(t, x) = \inf_{\pi} \mathbb{E}_{t,x} \left[\frac{\gamma}{2} \int_t^T \pi_s^2 ds + X_T^2 \right] ,$$

where $dX_t = -\pi_t dt + \sigma dW_t$. The HJB equation is

$$H_t + \inf_{\pi} \left(-\pi H_x + \frac{\sigma^2}{2} H_{xx} + \frac{\gamma}{2} \pi^2 \right) = 0 , \quad H(T, x) = x^2 .$$

The first-order condition gives

$$\pi^* = \frac{H_x}{\gamma} .$$

Thus

$$H_t + \frac{\sigma^2}{2} H_{xx} - \frac{1}{2\gamma} H_x^2 = 0 .$$

Using the ansatz

$$H(t, x) = a(t)x^2 + b(t) ,$$

we obtain

$$a'(t) - \frac{2}{\gamma} a(t)^2 = 0 , \quad b'(t) + \sigma^2 a(t) = 0 ,$$

with $a(T) = 1$ and $b(T) = 0$. We solve the equation for $a(t)$ as follows: write

$$\frac{a'(t)}{a^2(t)} = \frac{2}{\gamma} ,$$

then integrate w.r.t. t and then take reciprocal, $\frac{1}{a(t)} = -\frac{2}{\gamma}t + \text{constant}$, or

$$a(t) = \frac{\gamma}{-2t + C} ,$$

and where using $A(T) = 1$ we determine that $C = 2T + \gamma$. Solving for $b(t)$ is simply an integration of this solution for $a(t)$, that is $b(t) = \sigma^2 \int_t^T a(s) ds$. Hence

$$a(t) = \frac{\gamma}{\gamma + 2(T - t)}$$

$$b(t) = \frac{\gamma \sigma^2}{2} \log \left(1 + \frac{2(T - t)}{\gamma} \right) .$$

Therefore

$$H(t, x) = \frac{\gamma x^2}{\gamma + 2(T - t)} + \frac{\gamma \sigma^2}{2} \log \left(1 + \frac{2(T - t)}{\gamma} \right) ,$$

and the optimal control is

$$\pi_t^* = \frac{2X_t}{\gamma + 2(T - t)} .$$

2. Infinite-Horizon Log-Utility Consumption. The wealth process satisfies

$$dX_t = X_t (r + \pi_t(\mu - r)) dt + X_t \pi_t \sigma dW_t - c_t dt .$$

Let

$$H(x) = \sup_{\pi, c} \mathbb{E}_x \left[\int_0^\infty e^{-\rho t} \log(c_t) dt \right] .$$

The stationary HJB equation is

$$-\rho H(x) + \sup_{\pi, c \geq 0} \left(\log c + (rx + \pi x(\mu - r) - c)H_x + \frac{1}{2} \pi^2 x^2 \sigma^2 H_{xx} \right) .$$

Use the ansatz

$$H(x) = A + B \log x .$$

Then

$$H_x = \frac{B}{x} , \quad H_{xx} = -\frac{B}{x^2} .$$

The first-order condition with respect to consumption and the portfolio give

$$\begin{aligned} \frac{1}{c} - \frac{B}{x} &= 0 \\ B(\mu - r) - B\pi\sigma^2 &= 0 , \end{aligned}$$

and hence

$$\begin{aligned} c^* &= \frac{x}{B} \\ \pi^* &= \frac{\mu - r}{\sigma^2} \end{aligned}$$

Insert these optimal into the HJB, along with the ansatz, yields

$$-\rho A - \rho B \log(x) + \log(x) - \log(B) + rB + B \frac{(\mu - r)^2}{2\sigma^2} - 1 = 0 ,$$

wherein matching the coefficients of $\log x$ yields $B = \frac{1}{\rho}$. Therefore, the optimal consumption is

$$c_t^* = \rho X_t .$$

Matching non- $\log(x)$ terms yields $A = \frac{1}{\rho} \left(\log(\rho) + \frac{r}{\rho} + \frac{(\mu - r)^2}{2\rho\sigma^2} - 1 \right)$, and hence the value function is

$$H(x) = \frac{1}{\rho} \left(\log(x) + \log(\rho) + \frac{r}{\rho} + \frac{(\mu - r)^2}{2\rho\sigma^2} - 1 \right) .$$

3. Infinite Total Variation of Brownian Motion. Let

$$\Pi_n = \{0 = t_0 < t_1 < \cdots < t_n = T\}$$

be a sequence of partitions with $|\Pi_n| \rightarrow 0$ as $n \rightarrow \infty$, and write

$$\Delta W_i = W_{t_{i+1}} - W_{t_i} .$$

The total variation along Π_n is

$$V(W, \Pi_n) = \sum_{i=0}^{n-1} |\Delta W_i| .$$

By the inequality

$$\sum_{i=0}^{n-1} (\Delta W_i)^2 \leq \left(\max_i |\Delta W_i| \right) \sum_{i=0}^{n-1} |\Delta W_i| ,$$

we have

$$V(W, \Pi_n) \geq \frac{1}{\max_{i < n} |\Delta W_i|} \sum_{i=0}^{n-1} (\Delta W_i)^2 .$$

Almost surely,

$$\sum_{i=0}^{n-1} (\Delta W_i)^2 \longrightarrow T ,$$

by the quadratic variation of Brownian motion, while

$$\max_{i < n} |\Delta W_i| \longrightarrow 0 ,$$

by the continuity of Brownian paths. Therefore,

$$V(W, \Pi_n) \longrightarrow \infty \quad \text{a.s.} ,$$

and therefore,

$$\text{TV}_{[0,T]} := \lim_{n \rightarrow \infty} V(W, \Pi_n) = \infty \quad \text{almost surely} .$$